

Lecture 13

Geometric constructions in Extremal Combinatorics

Corner-free sets

Def: Corner in $\mathbb{Z}^2$: (x, y) $d \neq 0$
 $(x+d, y)$
 $(x, y+d)$



Q: How large a corner-free set
 $A \subseteq [N]^2$ can be?

Upper bound:

max size corner-free in $[N]^2 = f(N)$
 $f(N) = o(N^2)$ (reg. lemma)

Lower bound

- Behrend 3-AP-free const $c = 2\sqrt{2} \approx 2.828$
- Linial - Shraibman $(2+)$ 2.??
- Green $(2+)$ $c = 2\sqrt{2 \log_2 \frac{4}{3}} \approx 1.822...$

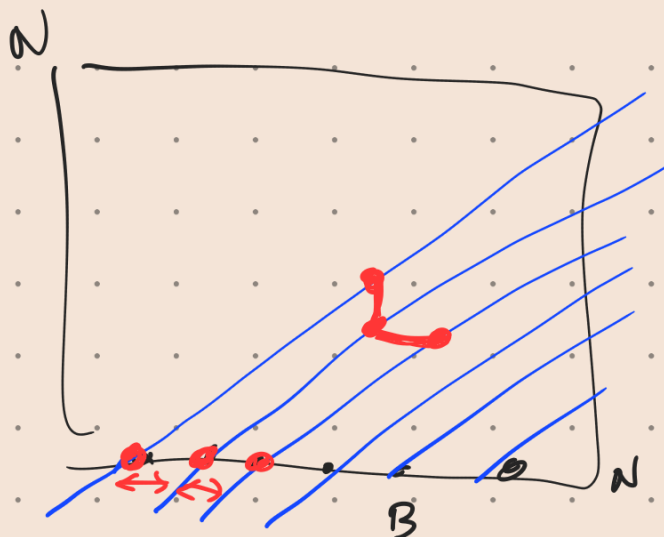
$$\frac{N^2}{2^{(c+o(1))\sqrt{\log N}}}$$

Rmk: Behrend: $B \subseteq [N]$ 3-AP-free $|B| = \frac{N}{2^{(2\sqrt{2}+o(1))\sqrt{\log N}}}$

$$A \subseteq [N]^2$$

Take $(x, y) \in A$
 if $x-y \in B$

$$|A| \leq |B| \cdot N$$



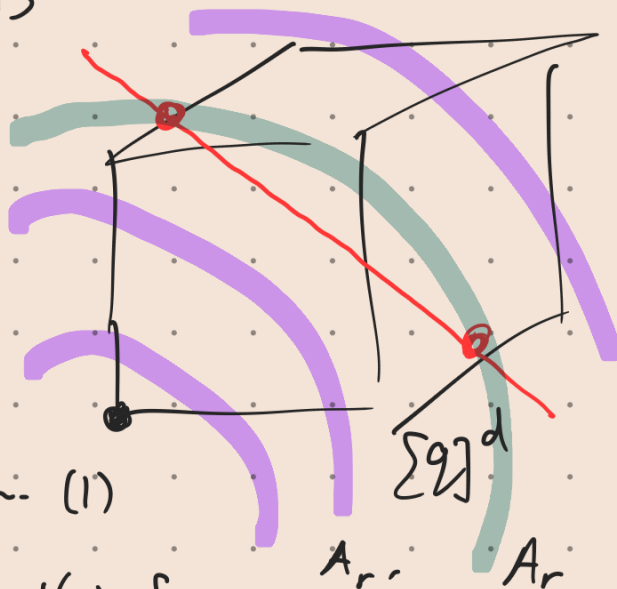
• Green's constr. for corner-free set. base q expansion.
 $N = q^d$ $x \in [N-1] \rightarrow x = \sum_{i=0}^{d-1} x_i \cdot q^i$

$$\pi : x \mapsto \pi(x) = (x_0, x_1, \dots, x_{d-1}) \in \mathbb{Z}^d$$

Idea

Geometric idea: any line $\cap$ any sphere at $\leq$ two pts

3 AP / corner $[N]^2 \iff$ line $[q]^d$



Consider set $A_r \subseteq [N-1]^2$

w/ (x, y) s.t.

$$\left\{ \begin{array}{l} \|\pi(x) - \pi(y)\|_2^2 = r \quad \dots (1) \\ \frac{q}{2} \leq x_i + y_i < \frac{3q}{2}, \quad \forall i \in \{0, \dots, d-1\} \end{array} \right.$$

$$\frac{q}{2} \leq x_i + y_i < \frac{3q}{2}, \quad \forall i \in \{0, \dots, d-1\}$$

Claim A_r is corner-free.

pf: Supp. $(x, y), (x+d, y), (x, y+d) \in A_r$

$$(1) \Rightarrow \|\pi(x) - \pi(y)\|_2^2 = \|\pi(x+d) - \pi(y)\|_2^2 = \|\pi(x) - \pi(y+d)\|_2^2 = r$$

$$\text{Claim (2)}: \dots \pi(x+d) + \pi(y) = \pi(x) + \pi(y+d)$$

Given (2), set $a = \pi(x) - \pi(y)$

$$b = \pi(x+d) - \pi(x) \stackrel{(2)}{=} \pi(y+d) - \pi(y)$$

$$(1) \Rightarrow \|a\|_2^2 = \|a+b\|_2^2 = \|a-b\|_2^2 = r$$

$\Rightarrow$ 3 pts $a, a+b, a-b$ on a line
all lie in the sphere of radius $\sqrt{r}$

Pf of (2): Induct on $i = 0, 1, \dots, d$ to show

$$(x+d)_i + y_i = x_i + (y+d)_i \quad i=0, 1, \dots$$

Supp. true for $\forall i < \hat{j}$, fix $\hat{j} \geq 0$.

• Write $x_{\geq \hat{j}} = \sum_{i \geq \hat{j}} x_i q^{i-\hat{j}}$ ($x_0, x_1, \dots, x_{\hat{j}}, x_{\hat{j}+1}, \dots$)

Ind. Hyp. & $(x+d)+y = x+(y+d)$

$$\Rightarrow (x+d)_{\geq \hat{j}} + y_{\geq \hat{j}} = x_{\geq \hat{j}} + (y+d)_{\geq \hat{j}}$$

$$\Rightarrow \underline{(x+d)_j + y_j = x_j + (y+d)_j \pmod{q}}$$

But both sides $\in [\frac{q}{2}, \frac{3q}{2})$ by def. of A_r

$$\Rightarrow (x+d)_j + y_j = x_j + (y+d)_j \quad \square$$

• Size of A_r , N^2 pts total

$$\#r \leq d \cdot q^2 \quad (x, y) \in A_r$$

For each $i \in [d]$, $\frac{q}{2} \leq x_i + y_i < \frac{3q}{2}$

$$\text{Size} \approx \left(\frac{3}{4}q^2 + O(q) \right)^d$$

$$\Rightarrow |A_r| \geq \frac{1}{dq^2} \left(\frac{3}{4}q^2 + O(q) \right)^d \quad \square$$



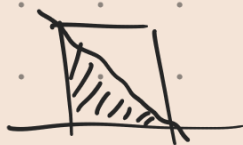
Rmk Behrend 3AP-free

$$N = q^d, \quad [q^d - 1] = [q]^d$$

$$x \rightarrow x_0, \dots, x_{d-1}$$

$$A_r = \{x \in [q^{d-1}] \text{ s.t.}$$

$$\begin{cases} \|\pi(x)\|_2^2 = r \\ 0 \leq x_i < q/2 \quad \forall i \end{cases}$$



_____ Erdős & Rothschild question on
edges in many triangles _____

Def: book is



edge = spine

book size = # triangles sharing the spine.

$$\text{Mantel} \Rightarrow \leq \frac{n^2}{4} \text{ edges} \\ \text{edge-density} < \frac{1}{2}$$

$\Rightarrow$ no Δ guaranteed
(book)

Edwards / Khadziivanov - Nikiforov



$$\forall n \times G \geq \frac{n^2}{4} + 1 \text{ edges} \Rightarrow \exists \frac{n}{6} \text{-size book}$$

Problem: What if every edge must be in a Δ ?

Def: $b(n, p) =$ largest book size in any
 $n \times G$ w./ $\geq p \binom{n}{2}$ edges & every edge
in $\geq 1 \Delta$.

Ques Erdős-Rothschild:

$$\forall p > 0, \quad b(n, p) \geq n^{\Omega(p)}?$$

Rmk : Removal lemma $\Rightarrow b(n, p) \geq e^{c \cdot (\log^* n)}$

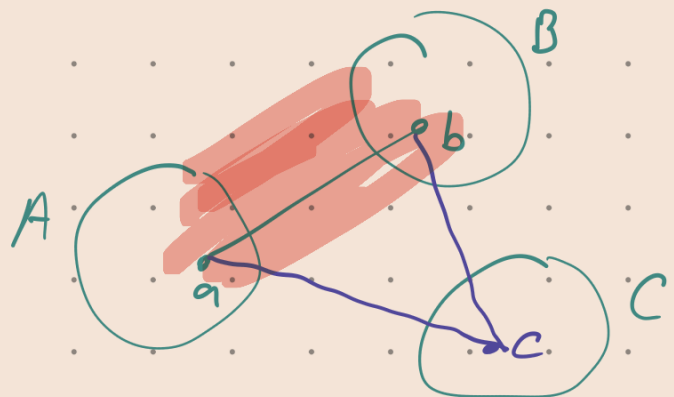
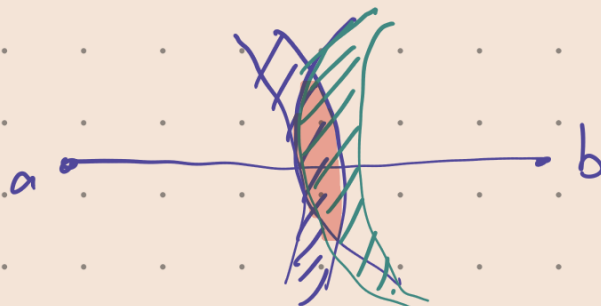
Thm (Fox-Loh 12) (not true up to edge density $\frac{1}{2} - o(1)$)

$$\left[\begin{array}{l} \exists G \text{ w/ } \frac{n^2}{4} (1 - e^{-(\log n)^{1/6}}) \text{ s.t.} \\ \forall \text{ edge } in \geq 1 \Delta \quad \& \quad b(G) \leq n^{\frac{14}{\log \log n}} \end{array} \right.$$

Idea : $\bullet$ in high-dim grid $[r]^d$, almost all pairs of pts are at some typical distance apart.

$$A = B = C = [r]^d$$

$a \sim b$ iff $\text{dist}(a, b) = \text{typical}$



Setup : $r \in \mathbb{N}$, $d = r^5$, $n = r^d$

$$A = B = C = [r]^d$$

(typical dist) $\mu = \frac{r^2 - 1}{6} \cdot d$

$$a \in A \sim b \in B \Leftrightarrow \|a - b\|_2^2 = \mu \pm d$$

$$a \in A \sim c \in C \Leftrightarrow \|a - c\|_2^2 = \frac{\mu}{4} \pm 2d$$

($\{B, c\} \cong \{A, c\}$)

Lem 1: $[A, B]$ almost complete around
 (almost all pairs of pts in $[r]^d$ are at μ typical dist)

✓ two unif chosen $X, Y \sim [r]^d$

$$\mathbb{P} \left(\|X - Y\|_2^2 = \mu \pm d \right) \geq 1 - 2e^{-d/2r^4}$$

pf: • $X = (X_1, \dots, X_d)$
 $Y = (Y_1, \dots, Y_d)$

$$\|X - Y\|_2^2 = \sum_i (X_i - Y_i)^2$$

• Fix $i \in [d]$, $\mathbb{E}(X_i - Y_i)^2 = \mathbb{E}(X_i^2 + Y_i^2 - 2X_i Y_i)$
 $X_i, Y_i \text{ indep}$
 $\sim [r]$ $= \mathbb{E}X_i^2 + \mathbb{E}Y_i^2 - 2\mathbb{E}X_i \mathbb{E}Y_i$
 $= 2(\mathbb{E}X_i^2 - (\mathbb{E}X_i)^2)$

$$\mathbb{E}X_i = \frac{r+1}{2}$$

$$\mathbb{E}X_i^2 = \frac{1}{r} \sum_{i=1}^r i^2 = \frac{(r+1)(2r+1)}{6}$$

$$\Rightarrow \mathbb{E}(X_i - Y_i)^2 = \frac{r^2 - 1}{6}$$

$$\Rightarrow \mathbb{E}\|X - Y\|_2^2 = \sum_{i=1}^d \mathbb{E}(X_i - Y_i)^2 = \mu$$

Use concentration of measure to conclude.

Def: $X(w)$ r.v. on d -dim prob. sp. $\Omega = \prod_{i=1}^d \Omega_i$

C -Lipschitz if $X(w)$ changes by $\leq C$
when one coord. of w is changed.

Hoeffding - Azuma Ineq X C -Lip. r.v. on
 d -dim, $\forall t \geq 0$

$$\mathbb{P}(|X - \mathbb{E}X| > t) \leq 2e^{-\frac{t^2}{2C^2d}}$$

$\|X - Y\|_2^2$ r.v. on dim- d , r^2 -Lip. $\blacksquare$

Left to show

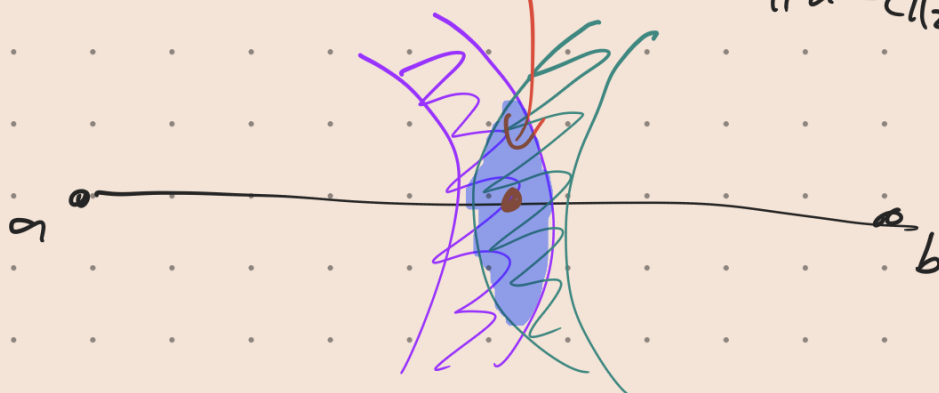
$n = r^d = r^{rs} = d^{d^{1/s}}$ { 1) $\forall$ ab edge $\in [A, B]$ is in
btr $[1, 15^d] \Delta \leq$
2) bootstrap edge-density to $\frac{1}{2} \pm o(1)$
blowup each u_x in $A \cup B$ to
 2^d copies

Pf of 1)

Lower bd
lattice
 $\geq 1/8^+$

$$\|a - b\|_2^2 = \mu \pm d$$

$$\|a - c\|_2^2 = \frac{\mu}{4} \pm 2d$$



Vpp bd.: NTS (blue area small)

Take $a \sim b$. $\|a-b\|_2^2 = \mu \pm d \dots (†)$

Set $x_i = b_i - a_i$

bd # half-attice pt c : $c_i = a_i + \frac{x_i}{2} + \frac{w_i}{2}$

where $w_i \in \mathbb{Z}$.

if $c \sim \frac{a+b}{2} \Rightarrow$

$\|c-a\|_2^2 = \|c-b\|_2^2 = \frac{\mu}{4} \pm 2d \dots (††)$

$$\hookrightarrow = \sum_i \left(\frac{x_i}{2} + \frac{w_i}{2} \right)^2 = \frac{\|b-a\|_2^2}{4} + \frac{1}{4} \sum w_i^2 + \frac{1}{2} \sum w_i x_i$$

$$\|c-b\|_2^2 = \sum_i \left(\dots \right)^2 = \dots - \frac{1}{2} \sum w_i x_i$$

$$\Rightarrow \|c-a\|_2^2 + \|c-b\|_2^2 = \frac{\|b-a\|_2^2}{2} + \frac{1}{2} \sum w_i^2$$

$$(†) \stackrel{(††)}{\Rightarrow} \sum w_i^2 \leq 9d$$

$$\# c = \# w \text{ s.t. } \sum w_i^2 \leq 9d$$

$$= \text{vol} \left(B_{\sqrt{3}d}^d \right) < 1.5^d. \quad \square$$